

CA2351: Ensure DataSet.ReadXml()'s input is trusted (code analysis)

CA2351: Ensure DataSet.ReadXml()'s input is trusted (code analysis) - dotpaul, learn.microsoft.com.

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Content

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CA2351: Ensure DataSet.ReadXml()'s input is trusted

Summarize this article for me

Property Value	Rule ID CA2351	Title Ensure DataSet.ReadXml()'s input is trusted	Category Security	Fix is breaking or non-breaking Non-breaking	Enabled by default in .NET 10 No	Applicable languages C# and Visual Basic
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Cause

The [DataSet.ReadXml](#) method was called or referenced, and not within autogenerated code.

This rule classifies autogenerated code b:

- Being inside a method named `ReadXmlSerializable`.
- The `ReadXmlSerializable` method has a [System.Diagnostics.DebuggerNonUserCodeAttribute](#).
- The `ReadXmlSerializable` method is within a type that has a [System.ComponentModel.DesignerCategoryAttribute](#).

[CA2361](#) is a similar rule, for when [DataSet.ReadXml](#) appears within autogenerated code.

Rule description

When deserializing a [DataSet](#) with untrusted input, an attacker can craft malicious input to perform a denial of service attack. There may be unknown remote code execution vulnerabilities.

For more information, see [DataSet and DataTable security guidance](#).

How to fix violations

- If possible, use [Entity Framework](#) rather than the [DataSet](#).
- Make the serialized data tamper-proof. After serialization, cryptographically sign the serialized data. Before deserialization, validate the cryptographic signature. Protect the cryptographic key from being disclosed and design for key rotations.

When to suppress warnings

It's safe to suppress a warning from this rule if:

- You know the input is trusted. Consider that your application's trust boundary and data flows may change over time.
- You've taken one of the precautions in [How to fix violations]().

Suppress a warning

If you just want to suppress a single violation, add preprocessor directives to your source file to disable and then re-enable the rule.

```
#pragma warning disable CA2351
// The code that's violating the rule is on this line.
#pragma warning restore CA2351
```

To disable the rule for a file, folder, or project, set its severity to `none` in the [configuration file](#).

```
[*.{cs,vb}]
dotnet_diagnostic.CA2351.severity = none
```

For more information, see [How to suppress code analysis warnings](#).

Pseudo-code examples

Violation

```
using System.Data;

public class ExampleClass
{
    public DataSet MyDeserialize(string untrustedXml)
    {
        DataSet dt = new DataSet();
        dt.ReadXml(untrustedXml);
    }
}
```

CA2350: Ensure DataTable.ReadXml()'s input is trusted

CA2352: Unsafe DataSet or DataTable in serializable type can be vulnerable to remote code execution attacks

CA2353: Unsafe DataSet or DataTable in serializable type

CA2354: Unsafe DataSet or DataTable in deserialized object graph can be vulnerable to remote code execution attack

CA2355: Unsafe DataSet or DataTable in deserialized object graph

CA2356: Unsafe DataSet or DataTable in web deserialized object graph

CA2361: Ensure autogenerated class containing DataSet.ReadXml() is not used with untrusted data

CA2362: Unsafe DataSet or DataTable in autogenerated serializable type can be vulnerable to remote code execution attacks